1. Display the current logged-in username, UID, primary GID and supplementary groups.

Command:

```
whoami
id
```

Output:

```
juhee
uid=1000(juhee) gid=1000(juhee) groups=1000(juhee),4(adm),27(sudo),100(users),

(Note: supplementary groups may differ on your Kali system.)
```

2. Display the account entry of a test user from /etc/passwd and identify the fields stored in the entry.

Command:

```
getent passwd labuser1
```

Output:

```
labuser1:x:1001:1001::/home/labuser1:/bin/sh

Fields: username : password placeholder : UID : GID : comment : home directory : login shell
```

3. Display the password/account status of a test user without displaying the actual password.

Command:

```
sudo passwd -S labuser1
```

Output:

```
labuser1 P 11/02/2026 0 99999 7 -1

P = password is set; L = account password is locked; NP = no password.
```

4. Create a new user named labuser1 using useradd and verify that the user has been created.

Command:

```
sudo useradd -m labuser1
getent passwd labuser1
id labuser1
```

Output:

```
labuser1:x:1001:1001::/home/labuser1:/bin/sh
uid=1001(labuser1) gid=1001(labuser1) groups=1001(labuser1)
```

5. Create labuser2 with a home directory and verify the home directory and its ownership.

Command:

```
sudo useradd -m labuser2
ls -ld /home/labuser2
```

Output:

```
drwx------ 2 labuser2 labuser2 4096 Nov  2 13:20 /home/labuser2
```

6. Create labuser3 with /bin/bash as the login shell and verify the assigned shell.

Command:

```
sudo useradd -m -s /bin/bash labuser3
getent passwd labuser3
```

Output:

```
labuser3:x:1003:1003::/home/labuser3:/bin/bash
```

7. Set a password for labuser1 and verify the password/account status.

Command:

```
sudo passwd labuser1
sudo passwd -S labuser1
```

Output:

```
New password: (type a password; characters are not shown)
Retype new password: (type it again)
passwd: password updated successfully
labuser1 P 11/02/2026 0 99999 7 -1
```

8. Change the password of labuser1 and verify the account status again.

Command:

```
sudo passwd labuser1
sudo passwd -S labuser1
```

Output:

```
New password: (enter the new password)
Retype new password: (enter it again)
passwd: password updated successfully
labuser1 P 11/02/2026 0 99999 7 -1
```

9. Use usermod to change labuser2's login shell to /bin/bash and verify the change.

Command:

```
sudo usermod -s /bin/bash labuser2
getent passwd labuser2
```

Output:

```
labuser2:x:1002:1002::/home/labuser2:/bin/bash
```

10. Change labuser2's home directory, first without moving contents and then with the option that moves the contents.

Command:

```
sudo touch /home/labuser2/sample.txt
sudo usermod -d /home/labuser2_new labuser2
getent passwd labuser2
sudo usermod -d /home/labuser2 labuser2
sudo usermod -d /home/labuser2_moved -m labuser2
ls -l /home/labuser2_moved
```

Output:

```
labuser2:x:1002:1002::/home/labuser2_new:/bin/bash
-rw-r--r-- 1 labuser2 labuser2 0 Nov  2 13:20 sample.txt

The first usermod changes the recorded home path only. The -m option moves the existing home contents.
```

11. Create a supplementary group named userlabgroup and add labuser1 to it using usermod. Verify membership.

Command:

```
sudo groupadd userlabgroup
sudo usermod -aG userlabgroup labuser1
id labuser1
groups labuser1
```

Output:

```
uid=1001(labuser1) gid=1001(labuser1) groups=1001(labuser1),1004(userlabgroup)
labuser1 : labuser1 userlabgroup
```

12. Lock labuser3 and verify that the account is locked. Then unlock it and verify the result.

Command:

```
sudo passwd -l labuser3
sudo passwd -S labuser3
sudo passwd -u labuser3
sudo passwd -S labuser3
```

Output:

```
passwd: password changed.
labuser3 L 11/02/2026 0 99999 7 -1
passwd: password changed.
labuser3 P 11/02/2026 0 99999 7 -1
```

13. Display detailed password-aging information for labuser3 using chage.

Command:

```
sudo chage -l labuser3
```

Output:

```
Last password change                                    : Nov 02, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
```

14. Set an account expiration date for labuser3 and verify the expiration information.

Command:

```
sudo chage -E 2026-12-31 labuser3
sudo chage -l labuser3
```

Output:

```
Account expires                                         : Dec 31, 2026

(Other password-aging fields remain unchanged.)
```

15. Modify labuser1 so that a specified existing group becomes the user's primary group. Verify the result.

Command:

```
sudo groupadd existinggroup
sudo usermod -g existinggroup labuser1
id labuser1
```

Output:

```
uid=1001(labuser1) gid=1005(existinggroup) groups=1005(existinggroup),1004(userlabgroup)
```

16. Display the users and groups associated with a selected account using id, groups and getent.

Command:

```
id labuser1
groups labuser1
getent group existinggroup
```

Output:

```
uid=1001(labuser1) gid=1005(existinggroup) groups=1005(existinggroup),1004(userlabgroup)
labuser1 : existinggroup userlabgroup
existinggroup:x:1005:
```

17. Delete labuser1 without removing its home directory. Verify the account entry and home directory separately.

Command:

```
sudo userdel labuser1
getent passwd labuser1
ls -ld /home/labuser1
```

Output:

```
(No account entry is returned by getent.)
drwx------ 2 1001 1001 4096 Nov  2 13:20 /home/labuser1

The account is deleted, but the home directory remains.
```

18. Delete labuser2 and remove its home directory using the appropriate userdel option. Verify the result.

Command:

```
sudo userdel -r labuser2
getent passwd labuser2
ls -ld /home/labuser2_moved
```

Output:

```
userdel: labuser2 mail spool (/var/mail/labuser2) not found

No account entry or home directory is returned. (The mail-spool notice may not appear on every system.)
```

19. Create a complete user-management workflow: create a user, set a password, assign a shell, add a supplementary group, lock/unlock the account and verify all settings.

Command:

```
sudo useradd -m -s /bin/bash labuser4
sudo passwd labuser4
sudo groupadd labteam
sudo usermod -aG labteam labuser4
sudo passwd -l labuser4
sudo passwd -S labuser4
sudo passwd -u labuser4
id labuser4
getent passwd labuser4
```

Output:

```
passwd: password updated successfully
passwd: password changed.
labuser4 L 11/02/2026 0 99999 7 -1
passwd: password changed.
uid=1006(labuser4) gid=1006(labuser4) groups=1006(labuser4),1007(labteam)
labuser4:x:1006:1006::/home/labuser4:/bin/bash
```

20. Clean up all test users and lab-only groups created during this lab. Verify that no unintended system account has been modified.

Command:

```
sudo userdel -r labuser3
sudo userdel -r labuser4
sudo groupdel userlabgroup
sudo groupdel existinggroup
sudo groupdel labteam
getent passwd labuser1 labuser2 labuser3 labuser4
getent group userlabgroup existinggroup labteam
```

Output:

```
All test-user and lab-only group entries have been removed.
(No output from the final getent commands means the entries are no longer present.)
```